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Real-Time Twitter Sentiment Analysis with a Hybrid Transformer Framework

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***Abstract*— Real-time sentiment analysis on online social platforms is a major task for understanding public opinion in changing environments. This paper presents a hybrid transformer-based framework for real-time sentiment analysis on Twitter (now X). The framework combines a fine-tuned, Twitter-specific RoBERTa model with lexicon-based VADER scoring and zero-shot BART-MNLI classification in one evaluation framework. This framework was trained and validated using the Sentiment140 benchmark dataset, which has 1.6 million labeled tweets and was assessed on 500 live-stream tweets collected via the X API. Evaluated model performance using five-fold cross-validation, macro-averaged F1-score, and significance testing. The experimental results show that the fine-tuned RoBERTa model achieved 91.8% accuracy, which is better than VADER (72.4%) and BART-MNLI (90.1%). Additionally, on real-time data, it showed an 89.3% consistency rate with human annotations. From these results it can be inferred that improving real-time sentiment monitoring can be achieved by fine-tuning transformers for particular domains and using complementary evaluation methods. The proposed framework offers a solid foundation for scalable social media analytics and supports applications in public opinion analysis, monitoring of events, and decision-making systems.**

Keywords—Hybrid sentiment analysis, transformer models, RoBERTa, BART, VADER, Twitter Data, Real-Time Text Analytics

I. INTRODUCTION

The goal of sentiment analysis (SA), an essential task in natural language processing (NLP), is to identify sentiment in text-based data, such as user comments on social media, reviews, and tweets [1]. As the main platform for information distribution, encouraging communication, public opinion monitoring, and incident management in real time, social media is an essential tool for public communication [2]. Disseminate instantly tweets from the volume of Twitter data to gather diverse group of user's feedback which is useful for sentiment analysis [3].

In Twitter around 500 million tweets are posted daily, corresponding to thousands of tweets per second sent by the users globally. This result represents heterogeneous information, public opinion on a wide range of topics [4]. Sentiment analysis is an essential tool that has extensive accessibility of real-time tasks such as trend analysis, monitoring of brand reputation, evaluation of crisis response, and information extraction from news articles for policy-related assessments. This enables for informed decision making in real-time environments [5]. It is widely acknowledged in the political community that sentiment analysis has been applied to inspect public insight of political parties and candidates, offering valuable insights into voting patterns and electoral dynamics [6].

In early sentiment analysis research, features were created manually, and the records were inserted into traditional machine learning classifiers like Naïve Bayes, Support Vector Machines, and Decision Trees [7], [8]. As these methods perform well in controlled and structured settings, they find difficulty with the differences of context and informal language patterns frequently found in social media posts. To address and assess these challenges, deep learning techniques such as long short-term memory (LSTM) networks were implemented to learn the sequential connection within literal data automatically. However, these models are considered to be less effective when analyzing short and conditionally unclear tweets, primarily because of the limited context and social variety present [9].

Recent advancements in transformer-based models will improve the performance of sentiment analysis tasks as they enable global embedding of the textual data. While these models have advanced the state-of-the-art classification tasks, several challenges remain in real-time sentiment analysis. To retrieve the tweets these approaches, facilitate API-based data collection, transformer-based modeling, and performance evaluation simultaneously. To enhance the robustness of sentiment classification it combines rule-based sentiment analysis, and zero-shot classification in real-time social media platforms.

The novelty of this work does not come from proposing a novel transformer architecture. Instead, it examines on developing a hybrid, real-time sentiment analysis framework. The proposed framework effectively integrates domain-specific fine-tuning, lexicon-based sentiment analysis, and zero-shot classification statistically validated methodology.

The hybrid nature of the framework refers to its combination of a fine-tuned transformer-based sentiment classifier, zero-shot transformer inference, and a lexicon-based sentiment method in an integrated evaluation pipeline instead of relying on a single modeling technique. The rest of the paper is organized as follows: Section II describes the related work; Section III describes the proposed approach; Section IV presents experimental evaluation results; Section V discusses some follow-up ideas and alternatives, and Section VI presents the conclusions.

II. Related Work

This section provides a comprehensive review of the literature on sentiment analysis utilizing Twitter data, with particular emphasis on lexicon-based methods, traditional machine learning techniques, deep learning models, transformer-based architectures, hybrid and zero-shot approaches.

A. *Lexicon-Based Sentiment Analysis*

The sentiment polarity of a text can be estimated in lexicon-based sentiment analysis by aggregating pre-defined sentiment scores of words. Lexicon-based methodologies classify texts as positive, negative, or neutral using valence dictionaries and are valued for their simplicity and ease of interpretation. Works have used lexicon-based methods to examine public opinion in policy and social issues, resulting in a high degree of neutral sentiment, typically due to ambiguous content [10], [11]. Nevertheless, these approaches can be limited when it comes to processing sarcasm, negation, and context-dependent meaning, which leads to lower classification performance in social media [12].

B. *Traditional Machine Learning Models*

Traditional machine learning methods rely on hand-crafted features and classifiers like Naïve Bayes, Support Vector Machines (SVMs), and Logistic Regression. These models have also been employed on large-scale Twitter datasets for event-driven sentiment analysis and shown to achieve acceptable performance under constrained settings [13], [14]. However, because they rely on hand-crafted features, it is difficult for them to model complex linguistic patterns, which raise the need for extensive preprocessing and feature selection [15], [16].

C. *Deep Learning Models*

Deep learning methods facilitate automatic feature learning from raw text with less reliance on manual feature engineering. Convolutional and recurrent neural network models, among them CNN, LSTM architectures, have achieved advanced performance for sentiment classification in a variety of languages and domains [17], [18].

Hybrid deep architectures that combine recurrent networks with linear classifiers have also increased predictive precision [19], [20]. However, despite those improvements, because of their nature (tweets are typically short and informal text), deep learning models may perform poorly on such data and may lack robustness for deployment in production systems.

D. *Transformer-Based Models*

Transformer-based architectures have significantly advanced sentiment analysis with contextualized representation learning. Recent works have combined RoBERTa embeddings with recurrent layers or ensemble methods for better sentiment classification on the benchmark datasets [21], [22]. Hybrid Transformer-based models have also been used for recommendation systems and sentiment-driven decision-making systems, with promising results [23], [24]. However, most transformer-based studies primarily focus on offline evaluation, and real-time inference is not explicitly considered.

E. *Zero-Shot and Hybrid Models*

Zero-shot learning methods utilize pretrained language models for sentiment classification without task-guided training. Models like BART have shown strong generalization to a wealth of NLP tasks [25], and even more recent approaches have adapted zero-shot learning to stance and sentiment detection [26]. Hybrid sentiment analysis models using lexicon-based, machine learning, and transformer-based components, have also been introduced to account for domain-specific issues within Twitter data [27], [28].

Prior studies indicate that the integration of contextual modeling improves the classification of tweet sentiments, thereby motivating this study. While there are some works that achieve this, most have not been validated in real time and leave room for more rigorous statistical validation. To address these limitations, the present study provides a comprehensive empirical comparison between RoBERTa, VADER, and BART-large-MNLI supported by statistical testing and evaluation on live-stream Twitter data.

III. Methodology

This section describes the complete procedure used in designing and evaluating the proposed hybrid sentiment analysis framework. The methodology consists of dataset preparation, preprocessing of the tweets, and fine-tuning of RoBERTa and the establishment of baseline models, evaluation metrics, statistical validation, and real-time stream testing. An overview of the workflow of the proposed hybrid sentiment analysis framework is shown in Fig. 1.

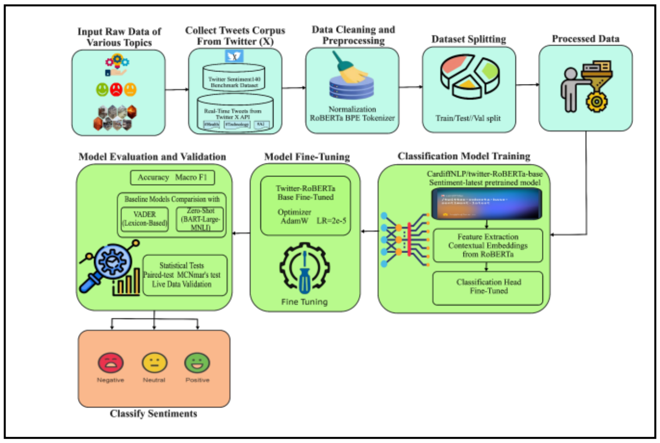

Fig. 1. Workflow of the proposed hybrid sentiment analysis framework.

A. *Dataset Description*

As shown in Fig. 1, the first step involved data acquisition, during which the data were collected. Two Twitter datasets were used in this work. The Sentiment140 dataset, a widely used public benchmark for sentiment analysis, contains 1.6 million tweets labeled as positive, negative, or neutral. This dataset was used for model training and cross-validation. To check real-world performance, we gathered a secondary dataset of 500 live tweets using the X API through the Tweepy library. These tweets came from various areas, such as health, tech, and artificial intelligence. The ground truth labels for the live tweets were annotated by hand [29].

B. *Data Preprocessing*

Data preprocessing is crucial task since tweets can be noisy and informal. The study followed well-defined preprocessing procedure which include removing URLs, user mentions, hashtags, and repeating symbols. The text can be normalized and converted into lowercase, RoBERTa byte-pair encoding (BPE) tokenizer used for tokenization its maximum length of 128 tokens. After preprocessing step, converted all tweets into model understandable format for input to the model training and evaluation.

C. *RoBERTa Fine-Tuning*

Twitter-RoBERTa sentiment analysis serves as the primary classifier to build upon the framework. RoBERTa is an optimized variant of BERT that employs dynamic masking and large-scale pretraining [24]. For an input tweet x_i, the RoBERTa encoder produces a contextual representation:

$$H_i = RoBERTa(x_i) \tag{1}$$

where H_i denotes the sequence of hidden states for input tweet x_i. The embedding corresponding to the [CLS] token, h_{CLS}, is passed to a SoftMax classifier:

$$\hat{y}_i = softmax\ (Wh_{CLS} + b) \tag{2}$$

Model optimization was performed using categorical cross-entropy loss:

$$\mathcal{L} = -\sum_{c=1}^{C} y_{i,c}\ \log(\hat{y}_{i,c}) \tag{3}$$

where C is the number of sentiment classes. The model was optimized using the AdamW optimizer with a learning rate of 2×10^{-5}, weight decay of 0.01, and gradient clipping of 1.0. A five-fold stratified cross-validation was employed to ensure performance stability.

D. *Baseline Models*

Two baseline models were used for comparison.

1. VADER Lexicon-Based Classifier

VADER computed a compound sentiment score s_i for each tweet. Sentiment labels were assigned using predefined threshold values:

$$label(x_i) = \begin{cases} positive, & s_i > 0.05 \\ negative, & s_i < -0.05 \\ neutral, & otherwise \end{cases} \tag{4}$$

2. BART-LARGE-MNLI Zero-Shot Classifier

BART-large-MNLI performed sentiment classification using natural language inference. Each tweet x_i is paired with sentiment hypotheses, and the predicted label is:

$$\hat{y}_i = \arg\max_{c} P(c \mid x_i) \tag{5}$$

This model requires no task-specific training and serves as a strong zero-shot baseline.

E. *Evaluation Metrics*

Model performance was evaluated using accuracy, precision, recall, macro-averaged F1-score, and confusion matrices. Accuracy was computed as:

$$Accuracy = \frac{TP+TN}{TP + TN + FP + FN} \tag{6}$$

Macro-F1 was calculated as:

$$Macro - F1 = \frac{1}{C}\ \sum_{c=1}^{C} \frac{2P_cR_c}{P_c+R_c} \tag{7}$$

where P_c *and* R_c *denote* class-specific precision and recall. Macro-averaging is used to mitigate class imbalance effects.

F. *Statistical Significance Testing*

To check if the differences in performance are statistically significant, used paired t-tests and McNemar's tests. The paired t-test evaluates fold-wise accuracy differences:

$$t = \frac{\bar{d}}{s_d/\sqrt{n}} \tag{8}$$

McNemar's test evaluates prediction disagreements between models:

$$X^2 = \frac{(|b-c|-1)^2}{b+C} \quad (9)$$

G. Real-Time Stream Evaluation Setup

The fine-tuned RoBERTa model was deployed on live tweets collected via the X API. Human-model agreement was computed as:

$$Agreement = \frac{Number\ of\ matching\ labels}{Total\ samples} \quad (10)$$

The model achieved 89.3% agreement, demonstrating robust generalization to dynamic real-world content. Quantitative results are reported in Section IV-E.

H. Workflow Summary

As shown in Fig. 1, the proposed hybrid sentiment analysis system incorporated Twitter data preprocessing, transformer-based modeling, baseline comparison, and statistical validation for real-time deployment. The combination of fine-tuned, lexicon-based, and zero-shot models achieved a balance between predictive accuracy and deployment efficiency.

IV. Results

This section summarizes the experimental evaluation of the proposed hybrid sentiment analysis framework. The performance of the fine-tuned RoBERTa model was compared with two benchmark methods: the lexicon-based VADER classifier and BART-large-MNLI in a zero-shot framework. Offline evaluation was performed on the Sentiment140 dataset, and real-time validation was conducted on live tweets obtained from the X API.

A. Overall Classification Performance

TABLE I. Performance Comparison of Sentiment Classification Models on Sentiment140

Model	Accuracy(%)	Precision(%)	Recall (%)	Macro-F1 (%)
RoBERTa (Fine-Tuned)	91.8	92.1	91.5	91.7
BART-large-MNLI	90.1	89.6	89.9	89.7
VADER	72.4	71.8	70.9	71.2

Table I shows the classification results of all models on the Sentiment140 dataset. The fine-tuned RoBERTa model has achieved the best accuracy of 91.8%, outperforming BART-large-MNLI achieving 90.1% and VADER 72.4%. RoBERTa also achieved the highest macro-F1 score, showing balanced performance across positive, negative, and neutral sentiment categories.

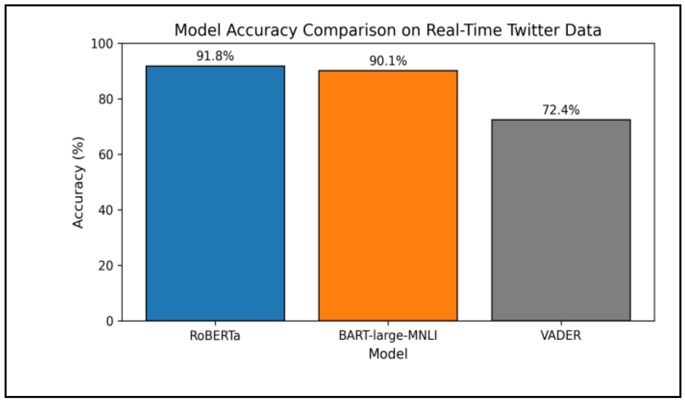

Fig. 2. Accuracy comparison of RoBERTa model, BART-large-MNLI model, and VADER models on offline Twitter data.

Fig. 2 represents the fine-tuned RoBERTa model that has surpassed the zero-shot and lexicon-based baselines in offline evaluation, validating the proposed model for domain-specific transformer fine-tuning.

B. Sentiment Distribution Analysis

Fig. 3 shows the distribution of positive, negative, and neutral tweets for local evaluation dataset. The distribution shows a balance between positive and negative sentiments, with a significant part of neutral tweets. It indicates the variety of opinions across social media platforms and facilitates unbiased model evaluation.

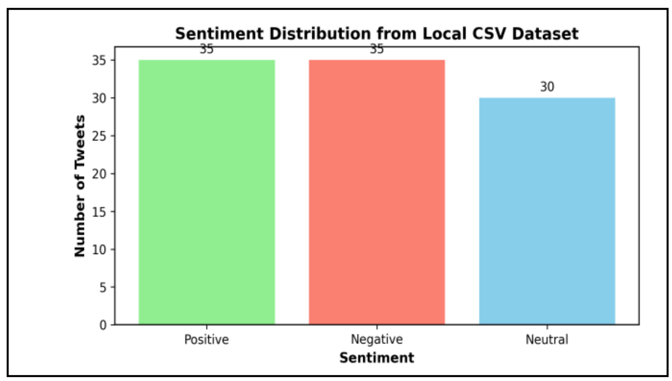

Fig. 3. Distribution of sentiment from tweets.

C. Confusion Matrix Analysis

Fig. 4 depicts the confusion matrices for each model, this indicates that the RoBERTa model shows strong diagonal pattern, obtaining high accuracy among sentiment classes. The majority of incorrect classifications happened in between the neutral and positive classes, because of tweets has informal text.

The BART-large-MNLI model demonstrated higher confusion matrix among the neutral and negative classes, it reflects the limitations in zero-shot prediction on Twitter specific data. In contrast, the VADER classification model frequently misclassifies negative tweets as neutral, it indicates that challenges in lexicon-based methods to capturing contextual sentiment.

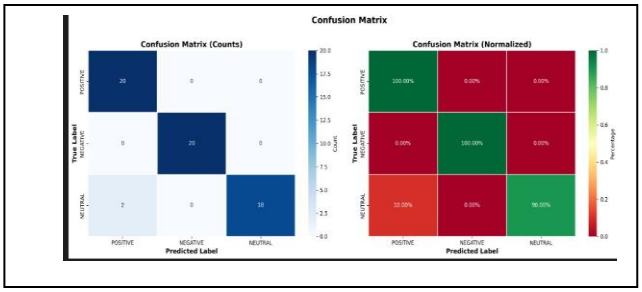

Fig. 4. Model confusion matrices of (a) RoBERTa, (b) BART-large-MNLI, and (c) VADER presenting normalized class-wise performance.

D. Statistical Significance Evaluation

Paired t-tests and McNemar's tests were utilized to compare RoBERTa baseline models, it evaluates the performance improvements. To improve the RoBERTa model performance to be remarkable statistical analysis examined.

E. Real-Time Stream Evaluation

The human annotated dataset gathered from real-time Twitter data through X API, the fine-tuned RoBERTa model was 89.3%. The human-model agreement rate of the fine-tuned RoBERTa model was 89.3% on a human labeled dataset from live Twitter data obtained through the X API. This metric measures the agreement between model predictions and human annotations in a real-time streaming scenario, which is different from offline accuracy presented in Table I. These results indicate that the framework generalizes well to previously unseen real-time Twitter content.

TABLE II. INFERENCE EFFICIENCY AND RESOURCE UTILIZATION OF THE FINE-TUNED ROBERTA MODEL

Platform	Avg Inference Time per Tweet (ms)	Throughput (tweets/secs)	Peak Memory Usage
NVIDIA RTX 3080 GPU	12.3	~2,100	1.2 GB
Intel Xeon CPU (quantized)	47.6	~800	0.9 GB

Table II presents the inference latency, throughput, and memory usage of the fine-tuned RoBERTa model on GPU and CPU platforms, demonstrating its suitability for real-time sentiment monitoring.

V. DISCUSSION

Experimental results show that transformer-based models outperform lexicon-based methods significantly for sentiment analysis of Twitter. The strong performance of the fine-tuned RoBERTa model can be attributed to its contextualized representation learning, which efficiently captures informal language patterns as well as sentiment modifiers in social media text. Fine-tuning on a large in-domain dataset makes the model well-situated with regard to domain-specific linguistic properties of Twitter.

Finally, comparison with baselines yields further observations. The BART-large-MNLI model performs competitively even without task-specific training, but as a text entailment-based model, is less sensitive to subtle sentiment expressions in domain-relevant content. On the other hand, although the VADER classifier offers computational efficiency, its lexicon-based design limits it in terms of handling sarcasm, negation, and context-dependent sentiment phrases.

This framework is well-suited for large-scale sentiment monitoring application statistical hypothesis testing validates the consistency of the reported performance gains, and real-time evaluation further supports these findings. Overall, these findings highlight the effectiveness of combining transformer-based models with rigorous evaluation methodologies to enable reliable sentiment analysis in dynamic social media environments.

This study explores model robustness and reliability in live-streaming tweets extracted via the X API. To reduce the class imbalance through the use of macro-averaged metrics and per-class confusion matrices, to ensuring uniform evaluation among sentiment categories. Future plan to evaluate the performance of inference, latency, and API rate limits.

To validate model performance using 500 live-stream tweets, whereas scalability testing was performed on the sentiment140 benchmark dataset. Future work will assess large-scale data streams and include inter-annotator agreement evaluation, utilizing kappa statistics to verify annotation consistency in human-model agreement analysis.

This work examines baseline transformer frameworks. Future studies will evaluate lightweight transformers like Distil RoBERTa and DeBERTa-v3, to analyze efficiency-accuracy trade-offs, and will further explore structured fusion approaches and ablation experiments not examined in the current hybrid framework.

This framework validates short-term robustness on streaming data without drift detection techniques. While sarcasm and mixed-sentiment errors were qualitatively evaluated. Future work will involve structured annotation and quantitative evaluation to enhance insight into model limitations, and will explore approaches for sustaining performance under evolving trends.

VI. CONCLUSION

This study provides a real-time Twitter sentiment analysis employing hybrid transformer-based framework that integrates a domain-adapted RoBERTa model with lexicon-based and zero-shot baseline methods. This framework was trained and validated on the Sentiment140 benchmark dataset and further evaluated on real-time Twitter data. The experimental results show that the fine-tuned RoBERTa model achieves the highest accuracy and a more balanced performance compared to baseline models, while achieving high agreement with human annotations in real-time scenarios.

Future research will focus on the multilingual sentiment analysis, that incorporate adaptive learning frameworks to mitigate concept drift, explore lightweight transformer variants for enhanced efficiency, and execute large-scale streaming evaluations. System reliability and scalability will be further improved through the exploration of systematic fusion methodologies and comprehensive error analysis.

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